Design Framework for Long-Term Physical Therapy Adherence

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Abstract

Maintaining long-term engagement in physical therapy remains a significant challenge, with studies suggesting that up to 50–70% of patients do not fully adhere to prescribed rehabilitation programs. This issue is particularly relevant for adult patients undergoing musculoskeletal rehabilitation, a process that often involves repetitive and time-intensive exercises that can lead to disengagement over time. While extended reality (XR) technologies have shown promise in improving engagement through immersive and interactive experiences, existing approaches often emphasize short-term interaction rather than sustained participation. Additionally, engagement is influenced not only by system features, but also by how users engage with these systems within real-world contexts over time.

The objective of this project is to develop a design framework for maintaining long-term engagement in physical therapy using XR systems. The framework focuses on rehabilitation contexts involving repetitive motor tasks and integrates principles from motivation theory, adaptive challenge design, feedback systems, and behavior change models to support sustained participation over time.

To achieve this objective, this work adopts a design science approach that synthesizes literature across human-computer interaction, rehabilitation science, and game design. These insights are organized into a conceptual framework consisting of four key components: motivation, interaction, engagement over time, and behavioral outcomes. The framework is supported by proposed evaluation metrics, including both experiential and behavioral measures, with validation and refinement reserved for future work.

The expected outcome is a theoretically grounded framework that can guide the design of XR-based rehabilitation systems aimed at improving long-term adherence. This work contributes to design research by integrating insights across disciplines and emphasizing engagement as a dynamic process shaped by system design, user behavior, and context. Additionally, this project supports future goals in developing human-centered, technology-driven solutions in healthcare that promote sustained engagement and behavior change over time.

Background and Motivation

Problem Context and Impact

Maintaining adherence to physical therapy programs remains a persistent challenge, with studies estimating that up to 50–70% of patients do not fully follow prescribed rehabilitation plans [3, 7]. This issue is particularly significant for adult patients undergoing musculoskeletal rehabilitation, especially athletes recovering from injury, as recovery in these cases often depends on consistent participation in repetitive and time-intensive exercises. When adherence is low, patients may experience prolonged recovery, reduced functional outcomes, and an increased risk of reinjury [23, 24].

While this work focuses on rehabilitation for athletes, similar challenges are observed across other populations, including older adults and individuals undergoing long-term recovery. In these contexts, maintaining motivation and consistency over time is critical yet difficult to sustain. These challenges highlight that rehabilitation is not only a clinical or physical process, but also a behavioral one, in which long-term engagement plays a central role in determining outcomes.

XR and Engagement in Rehabilitation

Recent research has explored the use of digital and interactive technologies, particularly extended reality (XR), to improve engagement in physical therapy. XR-based systems have been shown to increase both engagement and functional performance compared to traditional therapy methods [4, 16]. The immersive and interactive nature of XR allows users to receive real-time feedback, engage with dynamic environments, and experience a greater sense of presence during therapy [20].

In practice, XR-based systems may include features such as real-time visual feedback on movement accuracy, adaptive difficulty that adjusts exercise intensity based on user performance, and immersive environments that simulate task-based rehabilitation scenarios. For example, a patient recovering from a lower-body injury may perform exercises within a virtual environment that provides immediate feedback on posture while adjusting challenge levels to maintain engagement.

Insights from human-computer interaction and game design provide additional understanding of how engagement can be maintained. Self-Determination Theory (SDT) suggests that motivation is driven by the fulfillment of autonomy, competence, and relatedness, as illustrated in Fig. 1 [10, 25]. Similarly, gamification principles such as adaptive challenge, feedback, and progression systems are commonly used to maintain engagement in interactive systems [13, 15, 30].

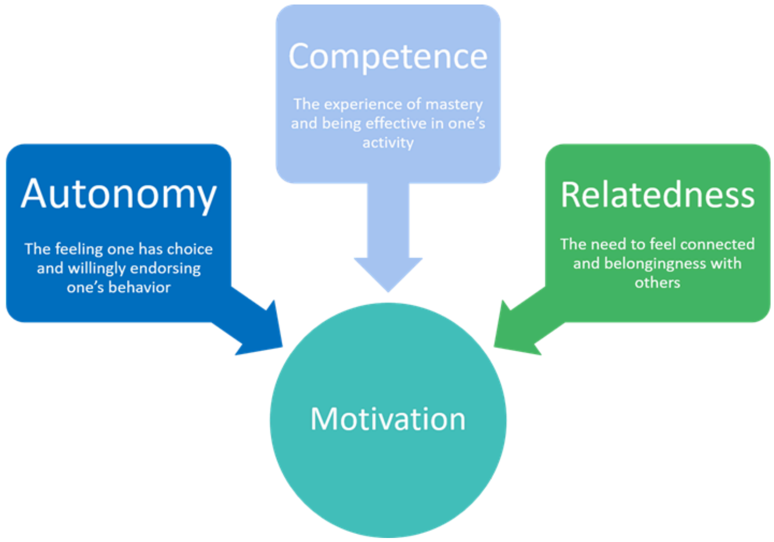

Fig. 1 Self-Determination Theory model showing autonomy, competence, and relatedness as drivers of motivation. Adapted from Przybylski et al. [10].

At the same time, engagement is influenced not only by system design, but also by broader behavioral and contextual factors. The Behavior Change Wheel, shown in Fig. 2, illustrates how behavior is shaped by the interaction of capability, opportunity, and motivation [29]. This highlights that sustaining engagement in rehabilitation requires consideration of not only system features, but also the user's environment, routines, and external constraints.

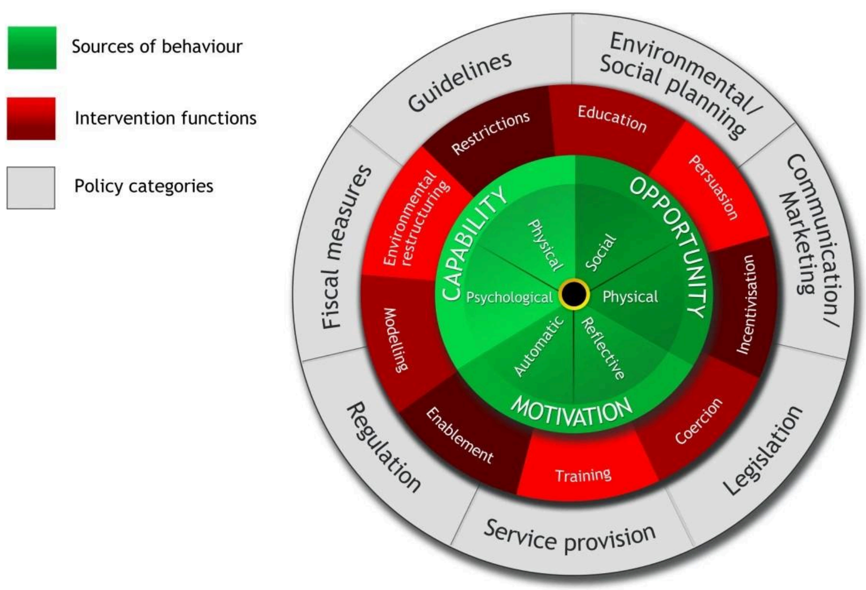

Figure 2. Behavior is influenced by interacting factors including motivation, capability, and opportunity. Adapted from Michie et al. [29].

Research Gaps and Motivation

Despite advances in XR technologies and engagement-focused design, there remains a lack of integrated approaches that address long-term engagement in rehabilitation. Existing research examines motivation, system design, and rehabilitation adherence as separate areas, rather than combining them into a cohesive framework. Additionally, many studies emphasize short-term engagement or system-level performance, with less attention given to long-term behavioral patterns such as adherence, disengagement, and re-engagement over time [21, 22].

This gap is especially important in rehabilitation contexts involving repetitive motor tasks, where sustained participation is key for successful outcomes. While concepts from motivation theory, gamification, and behavior change provide valuable insights, there is limited work that translates these ideas into structured design guidance for XR-based rehabilitation systems.

To address this gap, this project is informed by perspectives from design cognition and design methods, as well as beyond the artifact design. Rather than focusing solely on individual system features, this work considers how interaction design, user behavior, and contextual factors interact over time. This perspective supports the development of a framework that captures engagement as a dynamic process, providing a foundation for designing XR-based physical therapy systems that better support long-term adherence.

Design Objective

Scope and Context

The objective of this project is to develop a design framework for maintaining long-term engagement in physical therapy using extended reality (XR) systems. The framework focuses on adult patients undergoing musculoskeletal rehabilitation, with emphasis on athletes recovering from injury. These contexts often involve repetitive motor tasks that require consistent participation over extended periods, making sustained engagement critical for successful recovery outcomes [3, 7].

While the framework is developed with athlete rehabilitation as the primary focus, it is intended to be adaptable to other physical therapy populations and contexts with similar engagement challenges. For example, older adults and individuals undergoing long-term recovery, where adherence and motivation remain key concerns. By focusing on shared characteristics of rehabilitation, such as repetition, progression, and long-term commitment, the framework aims to balance specificity with broader applicability.

Framework Goals

The goal of the proposed framework is to support the design of XR-based rehabilitation systems that promote sustained engagement over time, rather than short-term interaction alone. To achieve this, the framework integrates principles from motivation theory, adaptive challenge design, feedback systems, and behavior change models.

The framework emphasizes the role of psychological needs such as autonomy and competence in driving motivation [10, 25], as well as the importance of adaptive challenge and feedback in maintaining engagement [13, 15]. Additionally, it incorporates behavior change perspectives that account for how engagement evolves over time and is influenced by contextual factors such as environment, routine, and user goals [29].

Instead of focusing solely on individual system features, the framework is designed to capture how these elements interact to influence user behavior over time. This includes not only initial engagement, but also patterns of sustained use, disengagement, and potential re-engagement.

Contribution and Intended Use

The proposed framework contributes to design research by providing a structured approach for integrating concepts from multiple domains into a cohesive model for long-term engagement in rehabilitation. Existing research examines motivation, interaction design, and adherence separately, but this framework aims to bring these elements together into a unified representation that can guide design decisions.

In practice, the framework can be used to inform the development of XR-based physical therapy systems by mapping design features, such as feedback mechanisms, adaptive difficulty, and progress tracking, to different components of the framework. It can also support evaluation by linking these components to measurable outcomes, including engagement, motivation, and adherence over time.

By framing engagement as a dynamic process influenced by both system design and contextual factors, this work extends beyond a single artifact and provides a foundation for future development, refinement, and empirical validation of XR-based rehabilitation systems.

Approach

Design Science Approach

This project adopts a design science research approach focused on the development of a conceptual framework as the primary design artifact. Design science emphasizes the creation of structured solutions to real-world problems through the integration of theoretical knowledge and design principles. In this context, the framework is intended to organize existing knowledge about motivation, interaction design, and behavior into a form that can guide the development of XR-based rehabilitation systems.

Rather than building a specific XR system, this work focuses on synthesizing insights from multiple domains, including rehabilitation science, human-computer interaction, game design, and behavior change research. This approach is appropriate because the challenge of maintaining long-term engagement in physical therapy is not due to a lack of individual techniques, but due to the absence of a cohesive structure that integrates these techniques into a unified design approach.

Framework Development

The framework is developed through a structured synthesis of literature and is organized into four primary components: motivation, interaction, engagement over time, and behavioral outcomes, as shown in Fig. 3. These components represent different aspects of how users interact with XR-based rehabilitation systems and how engagement is sustained over time.

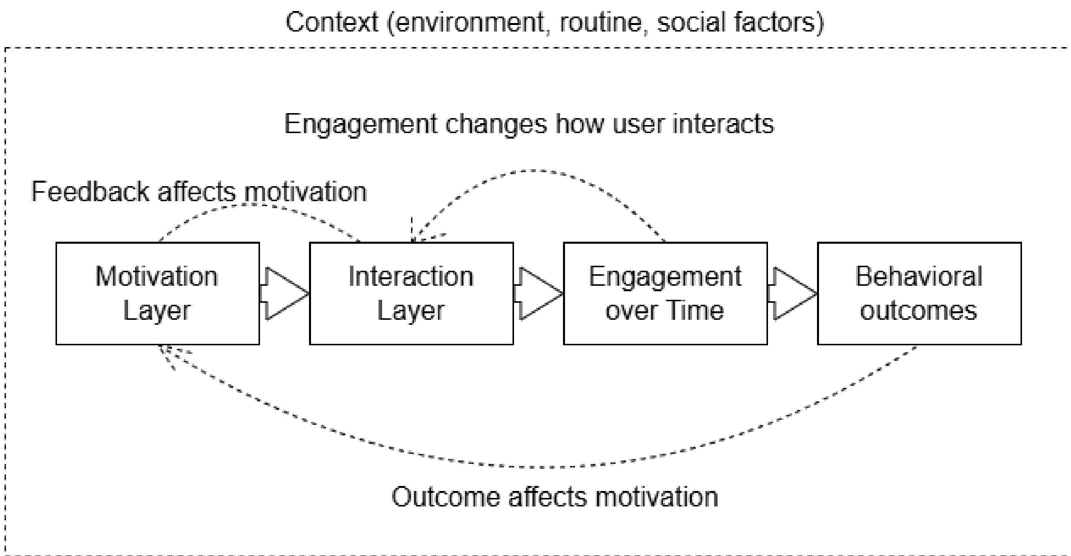

Fig. 3 Proposed framework for maintaining long-term engagement in XR-based physical therapy.

The motivation layer captures psychological drivers of engagement, including autonomy and competence, which influence a user's willingness to participate in rehabilitation activities [10, 25]. The interaction layer focuses on system-level design elements such as feedback, adaptive challenge, and progression, which shape the user's experience during each session [13, 15]. The engagement over time component reflects how user participation evolves across repeated interactions, including patterns of sustained use, disengagement, and re-engagement [21]. Finally, behavioral outcomes represent measurable results such as adherence, consistency of use, and completion of rehabilitation programs.

In addition to these components, the framework also accounts for contextual influences, including environment, routine, and social factors, which surround and affect all layers of the system. It also incorporates feedback loops between components. For example, behavioral outcomes can influence future motivation, while engagement over time can shape how users interact with the system. These relationships emphasize that engagement is not a linear, but a dynamic process influenced by multiple interacting factors.

In practice, this framework can be used to guide the design of XR-based rehabilitation systems by mapping specific features to different components. For example, real-time movement feedback can support the interaction layer, while adaptive difficulty can influence both interaction and motivation. This allows designers to systematically consider how design decisions impact both immediate user experience and long-term engagement.

Evaluation Strategy

To support future validation, the framework is paired with a set of proposed evaluation metrics that align with its key components. These include experiential metrics, such as engagement and perceived competence, as well as behavioral metrics, such as consistency of use, session frequency, and drop-off rates. Established tools such as the User Engagement Scale can be used to measure user experience, while constructs from Self-Determination Theory can be used to assess motivation [10, 12].

In practice, evaluation would be conducted through longitudinal studies in which participants use an XR-based rehabilitation system over an extended period. During these studies, system usage data would be collected to track behavioral patterns, while periodic surveys would assess changes in motivation and engagement. This approach allows for the evaluation of both short-term interaction quality and long-term adherence.

Due to the scope of this project, the framework is not implemented or empirically tested within this work. Instead, the focus is on developing a theoretically grounded and structured framework that can be applied and evaluated in future research. Future work will involve applying the framework to specific rehabilitation systems, conducting user studies, and refining the framework based on empirical findings.

Expected Outcomes and Implications

Expected Outcomes

The primary outcome of this work is the development of a theoretically grounded design framework for maintaining long-term engagement in XR-based physical therapy. This framework organizes key factors from the literature, such as motivation, adaptive challenge, feedback, and contextual influences, into a structured representation that can guide the design of rehabilitation systems.

By integrating insights from rehabilitation science, human-computer interaction, and game design, the framework provides a more cohesive understanding of how engagement can be sustained over time. It emphasizes engagement as a dynamic process that evolves across repeated interactions, rather than a static outcome measured at a single point in time. This perspective allows designers to consider not only how users initially interact with a system, but also how their behavior changes over the course of rehabilitation.

Research and Practical Implications

From a research perspective, this work contributes to design science by developing generalizable design knowledge in the form of a framework rather than a single system or prototype. It addresses a gap in existing literature by integrating disconnected areas of knowledge in motivation theory, XR interaction design, and rehabilitation adherence into a unified structure. In doing so, it aligns with approaches that extend beyond the artifact itself by considering how systems interact with user behavior and real-world contexts over time.

From a practical standpoint, the framework can be used to guide the design of XR-based rehabilitation systems by helping designers map specific features, such as feedback mechanisms, adaptive difficulty, and progression systems, to different components of engagement. This provides a structured approach for making design decisions that support both immediate user experience and long-term adherence.

In addition, the inclusion of proposed evaluation metrics offers a foundation for assessing engagement and adherence in future implementations. By combining experiential measures (perceived engagement and motivation) with behavioral data (consistency of use and drop-off rates), the framework supports a more comprehensive evaluation of rehabilitation technologies.

Limitations and Future Work

There are several limitations to this work. Most notably, the framework is not implemented or empirically evaluated within the scope of this project. As a result, its effectiveness in real-world settings is yet to be validated. Additionally, the framework is based on literature synthesis, which may not capture all contextual factors present in specific rehabilitation environments.

Another limitation lies in the adoption of XR-based rehabilitation systems in practice. Successful implementation requires buy-in from both clinical providers and patients. Healthcare providers may be hesitant to adopt new technologies due to concerns about cost, workflow disruption, or training requirements, while patients may be resistant to unfamiliar systems or changes to established rehabilitation routines. These factors highlight that the effectiveness of the framework depends not only on design quality, but also on user acceptance and integration within existing clinical practices.

Future work will focus on applying the framework to the design of XR-based rehabilitation systems and evaluating its effectiveness through user studies. This includes conducting longitudinal studies to assess how different design elements influence engagement and adherence over time. Further refinement of the framework may also involve adapting it to different user populations, rehabilitation types, and levels of physical ability, as well as incorporating insights from real-world implementation.

Future Goals

This project builds upon my current skillset and experience in design science by combining literature synthesis, framework development, and human-centered problem framing. I have developed experience analyzing complex problems through a design science perspective, identifying gaps in existing systems, and translating research insights into structured design opportunities through prior coursework. This project builds upon these skills by requiring the integration of knowledge from multiple domains, including rehabilitation science, human-computer interaction, motivation theory, and behavior change research. The emphasis on that effective design must take into account not only system features, but also the user needs, routines, and other contextual constraints over time also ties in nicely with design science.

In addition, this work also supports my future goals in developing human-centered, technology-driven solutions in healthcare and related domains. I am particularly interested in designing interactive systems that promote sustained engagement, behavior change, and improved long-term outcomes. This project strengthens my ability to work at the intersection of design, technology, and health by exploring how XR technologies can be applied to rehabilitation. The framework developed through this work also provides a foundation for future research involving the implementation and evaluation of digital health systems, while helping me grow as a designer who can approach problems from both a technical and systems-oriented perspective. Currently, I have plans to take this project a step further by proposing it as my Capstone project for my Master's degree.

AI Usage Acknowledgement

Grammarly, as well as minimal usage of Claude, were used to improve grammar, clarity, and readability in the drafting process. After using this tool, I carefully reviewed and edited the text and will take full responsibility for the final content.

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